Abnik Ahilasamy

AI Agent Evaluation Engineer | Enterprise Benchmarks, Tool Use & Agent Reliability
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Education

Arizona State University — Aug 2024 – May 2026
Master of Science in Data Science — *GPA: 3.97/4.0*

SRM Institute of Science and Technology — Aug 2020 – Jun 2024
Bachelor of Technology in Artificial Intelligence & Machine Learning — *GPA: 3.8/4.0*

Experience

CIPS-AI Labs — Jun 2026 – Present
Post-Training Researcher — *Arizona, USA*

- Built a reproducible, validation-checked legal-AI benchmark data pipeline for post-training evaluation across **873** Supreme Court land-dispute cases; defined **12** eval views across **8** sections and **4** precedent-sentiment categories.
- Generated **4.19M+ words** across **790** content-bearing records with **100% case-ID coverage** and **0 processing errors**; isolated **83** legitimately unsegmented cases as a tracked data-quality outcome for benchmark curation.
- Hardened long-running benchmark generation with a **5-case** smoke test, resumable checkpoints, validation gates, and bot-check recovery; supported a **26,688**-judgment corpus with a **4,039-node/7,082-edge** citation graph and **1,593** admitted land-dispute cases.

Larsen & Toubro — Apr 2024 – Jun 2024
LLM Engineer — *Chennai, IN*

- Built and benchmarked a production-oriented agentic RAG service with LangChain, LlamaIndex, FastAPI, and vector retrieval; optimized retrieval/inference with caching, batching, and memory optimization, reducing **p95 latency by 40%** and **hallucination rate by 35%** across a fixed **100-query** evaluation set.
- Defined faithfulness, groundedness, nDCG@10, and MRR benchmarks; added PII filters, constrained prompts, fallbacks, rollback gates, Prometheus, and OpenTelemetry tracing for failure analysis, reducing unsafe outputs by **50%** across **50** safety cases.

Additional Internships – ML & Data Analytics — Jun 2023 – Sep 2023
AI Engineer

- benchmarked **5+** NLP, time-series, and computer-vision projects with Python, Scikit-learn, TensorFlow, and Keras; compared LSTM/RNN/CNN validation, achieving **15–20%** improved sequence prediction and **90%+** CNN validation accuracy.

Projects

Enterprise-Bench v2 (DevRev Computer) | *Agent Evaluation, Benchmark Design, Harbor* — Sep 2026

- Contributed task design and evaluation feedback to DevRev Computer's **24-task**, **6-system** benchmark; piloted **5 UK L2 tasks** with GPT-5.6 Luna (**4/5 passed in one run**), analyzed a late-email failure, and proposed evaluation visualizations, sandboxing safeguards, and virtual-clock tests during livestream discussions.

SpotDiff Eval | *Python, Hugging Face, JSON, Benchmarking, VLMs* | GitHub — Aug 2026

- Built and open-sourced a multimodal evaluation harness over a **10-image/61**-difference benchmark; standardized JSON outputs and deterministic across **4** difference types with **0** ground-truth annotation files in model payloads.
- Benchmarked **7** vision models end-to-end and smoke-tested **8** additional models, publishing **3** reproducible development-leaderboard results and identifying **4** production failure modes; validated the CLI, Hugging Face dataset pipeline, schema checks, and regression suite with **19** automated tests, with gemini-3.1-flash scoring **32.8%** overall, **57.1%** precision, and **19.7%** recall.

Technical Skills

Evaluation & Benchmarking: LLM evals, agent benchmarks, Enterprise-Bench, benchmark task design, evaluation harnesses, deterministic scoring, regression testing, dataset curation, failure analysis, drift monitoring, safety testing
Agents & Inference: LangGraph, LangChain, LlamaIndex, RAG, MCP, Harbor, agent harnesses, tool use, sandboxing, data isolation, structured outputs, prompt engineering, p95 latency, caching, batching, memory optimization
Engineering: Python, FastAPI, SQL, Docker, Git, Linux, AWS, CI/CD, Prometheus, OpenTelemetry
Technical Content:**50K+ impressions**, **26K+ reach**, **600+ engagements** across LinkedIn and X
Publication: Brain Tumor Classification with Transfer Learning (ResNet-50, IEEE 2023)